Statistical Inference

Labix

February 9, 2026

Abstract

Notes for the basics of Probability Theory.

Contents

1 Fundamentals of Statistics **3**
1.1 Random Samples and Statistics . . . 3

2 Principles of Data Reduction **4**
2.1 Sufficiency Principle . . . 4
2.2 Likelihood Principle . . . 5
2.3 Conditionality Principle . . . 5
2.4 Equivariance Principle . . . 5

3 Estimators **6**
3.1 Finding Point Estimators . . . 6
3.2 Evaluating Point Estimators . . . 6
3.3 Finding Interval Estimators . . . 6

4 Hypothesis Testing **7**

1 Fundamentals of Statistics

1.1 Random Samples and Statistics

Definition 1.1.1 (Random Samples)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be random variables. We say that they are a random sample of size n from the population f if the following are true.

- $X_1, \dots, X_n$ are pairwise independent.
- The probability density (mass) function is given by $f_{X_i} = f_X$.

The fact that their probability density functions are equal imply that the random variables are identically distributed.

Definition 1.1.2 (Statistics)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. A statistic of the random sample is a random variable $T(X_1, \dots, X_n)$ for some function $T : \mathbb{R}^n \to \mathbb{R}$.

Definition 1.1.3 (Sample Mean)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Define the sample mean of the random sample to be the statistic defined by

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Definition 1.1.4 (Sample Variance)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Define the sample variance of the random sample to be the statistic defined by

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \overline{X})^2$$

Definition 1.1.5 (Sample Standard Deviation)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Define the sample standard deviation of the random sample to be the statistic defined by

$$S = \sqrt{S^2}$$

2 Principles of Data Reduction

2.1 Sufficiency Principle

Suppose that $(\Omega, \mathcal{F}, P)$ is a probability space. Often we do not know what P is actually like. To probe the properties of P, we may use statistics $T(X_1, \dots, X_n)$ on the probability space.

A sufficient statistic T is a statistic where, informally speaking, all information about P must come from T.

Definition 2.1.1 (Sufficient Statistics)
Let $(\Omega, \mathcal{F}, P_\theta)$ be a collection of probability spaces for $\theta \in I$. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let T be a statistic of the random sample. We say that T is sufficient if

$$P_{\theta_1}(X^{-1}(A) \mid T^{-1}(t)) = P_{\theta_2}(X^{-1}(A) \mid T^{-1}(t))$$

for all $\theta_1, \theta_2 \in I$ and all $A \in \mathcal{F}$ and all $t \in \mathbb{R}$.

Lemma 2.1.2
Let $(\Omega, \mathcal{F}, P_\theta)$ be a collection of probability spaces for $\theta \in I$. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let T be a statistic of the random sample. If the ratio

$$\frac{f_{X_1,\dots,X_n}(x_1, \dots, x_n)}{f_{T(X_1,\dots,X_n)}(x_1, \dots, x_n)}$$

does not depend on θ, then T is a sufficient statistics.

Proof
Firstly notice that if $t \notin \text{im}(T)$ then the probability is 0. Hence we only need to consider the case that $T(X_1, \dots, X_n) = T(x_1, \dots, x_n)$. Denote $\mathbf{X} = (X_1, \dots, X_n)$ and $\mathbf{x} = (x_1, \dots, x_n)$. We have that

$$\begin{aligned} P_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) &= \frac{P_\theta((\mathbf{X} = \mathbf{x}) \cap (T(\mathbf{X}) = T(\mathbf{x})))}{P_\theta(T(\mathbf{X}) = T(\mathbf{x}))} \\ &= \frac{P_\theta(\mathbf{X} = \mathbf{x})}{P_\theta(T(\mathbf{X}) = T(\mathbf{x}))} \\ &= \frac{f_{X_1,\dots,X_n}(x_1, \dots, x_n)}{f_{T(X_1,\dots,X_n)}(x_1, \dots, x_n)} \end{aligned}$$

so if the ratio does not depend on θ, we are done. ■

Example 2.1.3
Let $\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample of Bernoulli distribution with parameter $\theta \in (0, 1)$ (Note that for each θ, $P_{X_i} = f_{X_i}$ is different since the formula for f_{X_i} depends on θ). Then $T = \sum_{i=1}^n X_i$ is a sufficient statistic for θ.

Proof

Let $\theta \in \mathbb{R}$. We use the above lemma. We have

$$\begin{aligned} P_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) &= \frac{f_{X_1,\dots,X_n}(x_1,\dots,x_n)}{f_{T(X_1,\dots,X_n)}(x_1,\dots,x_n)} \\ &= \frac{\prod_{i=1}^n f_{X_i}(x_i)}{f_{X_1+\dots+X_n}(x_1,\dots,x_n)} \\ &= \frac{\prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i}}{\binom{n}{\sum_{i=1}^n x_i}\theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i}} \\ &= \frac{\theta^{\sum_{i=1}^n x_i}(1-\theta)^{\sum_{i=1}^n x_i}}{\binom{n}{\sum_{i=1}^n x_i}\theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i}} \\ &= \frac{1}{\binom{n}{\sum_{i=1}^n x_i}} \end{aligned}$$

■

Theorem 2.1.4 (Neyman-Fisher Factorization Theorem)
Let $(\Omega, \mathcal{F}, P_\theta)$ be a collection of probability spaces for $\theta \in I$. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let T be a statistic of the random sample. Let f_θ be the probability density function of P_θ. Then T is sufficient if and only if there exists probability density functions g_θ for each θ such that

$$f_\theta(x) = g_\theta(T(x))h(x)$$

for all $x \in \Omega$ and all $\theta \in I$.

Definition 2.1.5 (Minimal Sufficient Statistics)
Let $(\Omega, \mathcal{F}, P_\theta)$ be a collection of probability spaces for $\theta \in I$. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let T be a sufficient statistic of the random sample. We say that T is minimal if for any other sufficient statistic T', we have T is a function of T'.

Definition 2.1.6 (Ancillary Statistics)

Definition 2.1.7 (Complete Statistics)

Theorem 2.1.8 (Basu's Theorem)

2.2 Likelihood Principle

2.3 Conditionality Principle

2.4 Equivariance Principle

3 Estimators

Suppose some real life problem is modelling statistically by the probability space $(\Omega, \mathcal{F}, P)$. To probe what P looks like, we sample the space using random variables $X_1, \dots, X_n : \Omega \to \mathbb{R}$. Based on the sample, we may infer that P may have parameter $\widehat{\theta}$ (naturally $\widehat{\theta}$ should be a function of $X_1, \dots, X_n$). We may then need to evaluate how accurate $\widehat{\theta}$ is compared to the actual parameter θ.

There are two main branches in estimator theory: We need different methods for finding estimators, and we need to evaluate how accurate the estimators are.

There are also two types of estimators: Point estimators and interval estimators.

3.1 Finding Point Estimators

Definition 3.1.1 (Point Estimators)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n$ be a random sample. A point estimator of the random sample is a random variable $T(X_1, \dots, X_n)$ for some function $T : \mathbb{R}^n \to \mathbb{R}$.

Note: This the same definition of a statistic.

Definition 3.1.2 (Method of Moments)

Definition 3.1.3 (Maximum Likelihood Estimators)

3.2 Evaluating Point Estimators

Definition 3.2.1 (Bias of a Point Estimator)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let $\widehat{\theta}$ be a point estimator of the random sample. Define the bias of $\widehat{\theta}$ to be

$$\text{Bias}(\widehat{\theta}) = E[\widehat{\theta}] - \theta$$

Definition 3.2.2 (Unbiased Point Estimators)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let $\widehat{\theta}$ be a point estimator of the random sample. We say that $\widehat{\theta}$ is unbiased if $\text{Bias}(\widehat{\theta}) = 0$.

Definition 3.2.3 (Mean Squared Error)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let $\widehat{\theta}$ be a point estimator of the random sample. Define the mean squared error of $\widehat{\theta}$ to be

$$\text{MSE}(\widehat{\theta}) = E[(\widehat{\theta} - \theta)^2]$$

Theorem 3.2.4 (The Bias Variance Trade Off)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let $\widehat{\theta}$ be a point estimator of the random sample. Then we have

$$\text{MSE}(\widehat{\theta}) = \text{Var}(\widehat{\theta}) + (\text{Bias}(\widehat{\theta}))^2$$

3.3 Finding Interval Estimators

4 Hypothesis Testing

5 Regression Analysis

Definition 5.0.1 (Linear Regression Space)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ be a random sample. Define the linear regression space of the sample to be

$$\mathcal{H}(X) = \left\{ \sum_{i=1}^{n} \beta_i X_i + \alpha \mid \alpha, \beta_i \in \mathbb{R} \right\}$$

Definition 5.0.2 (Linear Regression)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ be a random sample. Let $Y : \Omega \to \mathbb{R}^n$ be a random sample. Let $L : \mathcal{H} \to \mathbb{R}$ be a loss function. The linear regression of Y on X with L is a random variable $\widehat{Y} \in \mathcal{H}(X)$ such that

$$L(\widehat{Y}) = \min\{L(Z) \mid Z \in \mathcal{H}(X)\}$$

Proposition 5.0.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ be a random sample. Let $Y : \Omega \to \mathbb{R}^n$ be a random sample. Then the linear regression of Y on X with the mean squared error exists and is unique.